

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****1.1. Product identification**

Product Code/Catalogue Number: 981380  
SDS Number: D14832\_SDS\_Alpha-HBDH \_EN  
Product Name: **Alpha-HBDH**

**1.2. Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use: In vitro diagnostic.  
Uses advised against: No Information available

**1.3. Details of the supplier of the safety data sheet**

Company: **Thermo Fisher Scientific Oy**  
Ratastie 2,  
FI-01620 Vantaa, Finland  
Telephone number: +358 10 329200  
E-mail address: system.support.fi@thermofisher.com

**1.4. Emergency telephone number**

CHEMTREC INTERNATIONAL +1 703-741-5970

**SECTION 2: HAZARDS IDENTIFICATION****2.1. Classification of the substance or mixture****CLP Classification - Regulation (EC) No 1272/2008**

Based on available data, the classification criteria are not met

**2.2. Label elements**

None required

**2.3. Other hazards**

No information available

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS****3.2. Mixtures**

| Component                           | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                     |
|-------------------------------------|----------|---------------------------------------------------------------------------------------|
| Sodium azide<br>(CAS #: 26628-22-8) | < 0.1 %  | Acute Tox. 2 (H300)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410)<br>(EUH032) |

| Component    | Reach Registration Number |  |
|--------------|---------------------------|--|
| Sodium azide | 01-2119457019-37-xxxx     |  |

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Full text of Hazard Statements: see section 16

#### **SECTION 4: FIRST AID MEASURES**

##### **4.1. Description of first aid measures**

###### **General Advice**

If symptoms persist, call a physician.

###### **Inhalation**

Move to fresh air. If not breathing, give artificial respiration. Consult a physician.

###### **Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.

###### **Eye Contact**

Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

###### **Ingestion**

Clean mouth with water and drink afterwards plenty of water.

##### **4.2. Most important symptoms and effects, both acute and delayed**

No information available.

##### **4.3. Indication of any immediate medical attention and special treatment needed**

Treat symptomatically.

#### **SECTION 5: FIREFIGHTING MEASURES**

##### **5.1. Extinguishing media**

###### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

###### **Extinguishing media which must not be used for safety reasons**

No information available.

##### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

###### **Hazardous Combustion Products**

None under normal use conditions.

##### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### **SECTION 6: ACCIDENTAL RELEASE MEASURES**

##### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment. Ensure adequate ventilation.

##### **6.2. Environmental precautions**

Prevent further leakage or spillage if safe to do so. Prevent entry into waterways, sewers, basements or confined areas.

##### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material.

##### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7: HANDLING AND STORAGE****7.1. Precautions for safe handling**

Ensure adequate ventilation. Avoid contact with skin and eyes.

**7.2. Conditions for safe storage, including any incompatibilities**

Keep container tightly closed in a dry and well-ventilated place.

**7.3. Specific end use(s)**

Use in laboratories

**SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION****8.1. Control parameters****Component Exposure Limits**

| Component    | Finland                                                                                 | European Union                                                  | The United Kingdom                                              | Germany                               |
|--------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------|
| Sodium azide | TWA: 0.1 mg/m <sup>3</sup> 8 tunteina<br>STEL: 0.3 mg/m <sup>3</sup> 15 minuutteina lho | Skin<br>TWA 0.1 mg/m <sup>3</sup><br>STEL 0.3 mg/m <sup>3</sup> | Skin<br>TWA 0.1 mg/m <sup>3</sup><br>STEL 0.3 mg/m <sup>3</sup> | MAK 0.2 mg/m <sup>3</sup> (inhalable) |

| Component    | Sweden                                                                                        | Norway                                                                                                   | Denmark                                   | France                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Sodium azide | Binding STEL: 0.3 mg/m <sup>3</sup> 15 minuter<br>TLV: 0.1 mg/m <sup>3</sup> 8 timmar.<br>NGV | TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>STEL: 0.3 mg/m <sup>3</sup> 15 minutter. value from the regulation | TWA: 0.1 mg/m <sup>3</sup> 8 timer<br>Hud | TWA / VME: 0.1 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 0.3 mg/m <sup>3</sup> . restrictive limit<br>Peau |

**8.2. Exposure controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

**Personal protective equipment****Eye Protection**

Safety glasses with side-shields (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or

other symptoms are experienced.  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

**SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES****9.1. Information on basic physical and chemical properties**

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | No information available |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No data available        |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No data available        |                                          |
| <b>Flash Point</b>                             | No data available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |

**9.2. Other information**

No data available

**SECTION 10: STABILITY AND REACTIVITY****10.1. Reactivity**

No data available

**10.2. Chemical stability**

Stable under normal conditions

**10.3. Possibility of hazardous reactions**

No information available.

**10.4. Conditions to avoid**

None known.

**10.5. Incompatible materials**

Heavy metals.

**10.6. Hazardous decomposition products**

None under normal use conditions.

**SECTION 11: TOXICOLOGICAL INFORMATION****11.1. Information on toxicological effects****Product Information**

No acute toxicity information is available for this product

**(a) acute toxicity;****Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component    | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|--------------|-------------------------|-------------|-----------------|
| Sodium azide | LD50 = 27 mg/kg ( Rat ) | -           |                 |

**(b) skin corrosion/irritation;**

No data available.

**(c) serious eye damage/irritation;**

No data available.

**(d) respiratory or skin sensitization;****Respiratory**

No data available.

**Skin**

No data available.

**(e) germ cell mutagenicity;**

No data available

**(f) carcinogenicity;**

No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

No data available.

**(h) STOT-single exposure;**

No data available.

**(i) STOT-repeated exposure;**

No data available.

**Target Organs**

No information available.

**(j) aspiration hazard;**

No data available.

**Symptoms / effects, both acute and delayed**

No information available

**SECTION 12: ECOLOGICAL INFORMATION**

**12.1. Toxicity**

| Component    | Freshwater Fish                                                                                                                                                     | Water Flea | Freshwater Algae | Microtox |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|----------|
| Sodium azide | LC50: = 5.46 mg/L, 96h<br>flow-through<br>(Pimephales promelas)<br>LC50: = 0.7 mg/L, 96h<br>(Lepomis macrochirus)<br>LC50: = 0.8 mg/L, 96h<br>(Oncorhynchus mykiss) |            |                  |          |

**12.2. Persistence and degradability**

No information available

**12.3. Bioaccumulative potential**

No information available

**12.4. Mobility in soil**

No information available

**12.5. Results of PBT and vPvB assessment**

No data available for assessment.

**12.6. Other adverse effects**

None known

**SECTION 13: DISPOSAL CONSIDERATIONS**

**13.1. Waste treatment methods**

**Waste from Residues / Unused Products**

Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of in accordance with local regulations.

**SECTION 14: TRANSPORT INFORMATION**

|                                  | IMDG/IMO      | ADR           | IATA          |
|----------------------------------|---------------|---------------|---------------|
|                                  | Not regulated | Not regulated | Not regulated |
| 14.1. UN number                  | -             | -             | -             |
| 14.2. UN proper shipping name    | -             | -             | -             |
| 14.3. Transport hazard class(es) | -             | -             | -             |
| 14.4. Packing group              | -             | -             | -             |

**14.5. Environmental hazards**

No hazards identified

**14.6. Special precautions for user**

No special precautions required

**14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code**

Not applicable, packaged goods

**SECTION 15: REGULATORY INFORMATION**

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

International Inventories X = listed

| Component    | EINECS    | ELINCS | NLP | TSCA | DSL | NDSL | PICCS | ENCS | IECSC | AICS | KECL         |
|--------------|-----------|--------|-----|------|-----|------|-------|------|-------|------|--------------|
| Sodium azide | 247-852-1 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-3135<br>7 |

**National Regulations**

| Component    | Germany - Water Classification (VwVwS) | Germany - TA-Luft Class |
|--------------|----------------------------------------|-------------------------|
| Sodium azide | WGK 2                                  |                         |

**15.2. Chemical safety assessment**

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

**SECTION 16: OTHER INFORMATION****Full text of H-Statements referred to under sections 2 and 3**

H300 - Fatal if swallowed

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

EUH032 - Contact with acids liberates very toxic gas

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

# SAFETY DATA SHEET

Alpha-HBDH

Revision Date 15-Mar-2019

Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - Volatile Organic Compounds

## Key literature references and sources for data

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

|                            |                                    |
|----------------------------|------------------------------------|
| <b>Version</b>             | 2                                  |
| <b>Revision Date</b>       | 15-Mar-2019                        |
| <b>Reason for revision</b> | SDS section(s) updated:, 1, 3, 16. |

## Disclaimer

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